

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the October/November 2007 question paper

9702 PHYSICS

9702/02

Paper 2 (AS Structured Questions), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

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- 1 (a) systematic: e.g. constant error (in all readings)
cannot be eliminated by averaging
error in measuring instrument
random: e.g. readings scattered (equally) about true value
error due to observer
can be eliminated by averaging
(only if averaging not included for systematic)
- B1
B1 [2]
- (b) $15 = \pi \times R^2 \times 20$
 $R = 0.4886 \text{ cm}$ (accept any number of s.f.)
 % uncertainty in $V = 3.3 \%$ (or $0.5/15$)
 % uncertainty in $L = 0.5 \%$ (or $0.1/20$)
 % uncertainty in $R = 1.9 \%$ (i.e. one half of the sum)
 $R = 0.489 \pm 0.009 \text{ cm}$
- C1
C1
C1
C1
A1 [5]
- 2 (a) $3.5 T$
- B1 [1]
- (b) (i) distance = average speed $\times$ time (however expressed)
 $= 14 \text{ m}$
- C1
A1 [2]
- (ii) distance = $5.6 \times (T - 5)$ (or $3.5T - 14$)
- A1 [1]
- (c) $3.5T = 14 + 5.6(T - 5)$
 $T = 6.7 \text{ s}$
- C1
A1 [2]
- (d) (i) acceleration = $(5.6 / 5) = 1.12 \text{ m s}^{-2}$
 force = ma
 $= 75 \text{ N}$
- C1
C1
A1 [3]
- (ii) power = (force $\times$ speed) = $\{75 + 23\} \times 4.5$
 $= 440 \text{ W}$
 (allow $1/2$ for 234 W , $0/2$ for 338 W or 104 W)
- C1
A1 [2]
- 3 (a) (i) potential energy: stored energy available to do work
- B1 [1]
- (ii) gravitational: due to height/position of mass OR distance from mass
 OR moving mass from one point to another
 elastic: due to deformation/stretching/compressing
- B1
B1 [2]
- (b) (i) height raised = $(61 - \{61 \cos 18\}) = 3.0 \text{ cm}$
 energy = $(mgh = 0.051 \times 9.8 \times 0.030) = 1.5 \times 10^{-2} \text{ J}$
- C1
A1 [2]
- (ii) moment = force $\times$ perpendicular distance
 $= 0.051 \times 9.8 \times 0.61 \times \sin 18$
 $= 0.094 \text{ N m}$
- C1
A1 [2]

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- 4 (a) brittle B1 [1]
- (b) Young modulus = stress / strain
 $= (9.5 \times 10^8) / 0.013$
 $= 7.3 \times 10^{10} \text{ Pa}$ (allow $\pm 0.1 \times 10^{10} \text{ Pa}$) C1
A1 [2]
- (c) stress = force / area C1
(minimum) area $= (1.9 \times 10^3) / (9.5 \times 10^8)$
 $= 2.0 \times 10^{-6} \text{ m}^2$ C1
(max) area of cross-section $= (3.2 - 2.0) \times 10^{-6}$
 $= 1.2 \times 10^{-6} \text{ m}^2$ A1 [3]
- (d) when bent, 'top' and 'bottom' edges have different extensions M1
with thick rod, difference is greater (than with a thin rod) A1
so breaks with less bending A0 [2]
- 5 (a) amplitude between 6.5 squares and 7.5 squares on 3 peaks B2
(allow 1 mark if outside this range but between 6.0 and 8.0 squares)
correct phase (ignore lead/lag, look at x-axis only and allow $\pm 1/2$ square) B1 [3]
- (b) $\lambda = ax / D$ C1
 $540 \times 10^{-9} = (0.700 \times 10^{-3} x) / 2.75$ C1
 $x = 2.12 \text{ mm}$ A1 [3]
- (c) (i) same separation B1
bright areas brighter (1)
dark areas, no change (1)
(allow 'contrast greater' for 1 mark if dark/light areas not discussed)
fewer fringes observed (1) any two, 1 each B2 [3]
- (ii) smaller separation of fringes B1
no change in brightness B1 [2]
- 6 (a) power = VI C1
current $= 10.5 \times 103 / 230$ M1
 $= 45.7 \text{ A}$ A0 [2]
- (b) (i) p.d. across cable = 5.0 V C1
 $R = 5.0 / 46$ C1
 $= 0.11 \Omega$ A1 [3]
- (ii) $R = \rho L / A$ C1
 $0.11 = (1.8 \times 10^{-8} \times 16 \times 2) / A$ C1
 $A = 5.3 \times 10^{-6} \text{ m}^2$ A1 [3]
(wires in parallel, not series, allow max 1/3 marks)

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(c) (i) either power = V^2 / R or power $\propto V^2$ C1
ratio = $(210 / 230)^2 = 0.83$ A1 [2]

(ii) resistance of cable is greater M1
greater power loss/fire hazard/insulation may melt
wire may melt/cable gets hot A1 [2]

7 (a) most α -particles deviated through small angles B1
(accept 'undeviated')
few α -particles deviated through angles greater than 90° B1 [2]

(b) (i) allow $10^{-9} \text{ m} \rightarrow 10^{-11} \text{ m}$ B1 [1]

(ii) allow $10^{-13} \text{ m} \rightarrow 10^{-15} \text{ m}$ B1 [1]
(if (i) and (ii) out of range but (ii) = 10^{-4} (i), then allow 1 mark)
(if no units or wrong units but (ii) = 10^{-4} (i), then allow 1 mark)